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Tavapadon as Adjunctive Treatment for Parkinson Disease
The TEMPO-3 Randomized Clinical Trial

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IMPORTANCE Development of motor fluctuations is common in people with Parkinson disease (PD) treated with oral levodopa, and currently available dopamine (D) agonists adjunctive to levodopa offer additional motor control but may increase risk of adverse events (AEs) via preferential activation of D2/D3 receptors. Tavapadon is a novel, investigational, oral, once-daily, selective D1/D5 agonist that may improve motor control while minimizing AEs commonly associated with D2/D3 receptor activation.

OBJECTIVE To evaluate the efficacy, safety, and tolerability of tavapadon as adjunctive therapy to oral levodopa in adults with PD experiencing motor fluctuations.

DESIGN, SETTING, AND PARTICIPANTS This was a phase 3, double-blind, placebo-controlled randomized clinical trial conducted between September 2020 and February 2024 with a 4-week safety follow-up. Participants were recruited from 148 sites across 14 countries. Participants were enrolled after screening adults with PD who were experiencing fluctuations while receiving stable oral levodopa (≥400 mg daily).

INTERVENTIONS Participants were randomized 1:1 to flexible-dose tavapadon (5-15 mg once daily) or placebo adjunctive to oral levodopa for 27 weeks.

MAIN OUTCOMES AND MEASURES The primary end point was change from baseline to week 26 in total daily on-time (ie, good mobility, smoother movement, fewer motor/nonmotor symptoms) without troublesome dyskinesia (referred to as *good-on-time*). The key secondary end point was change from baseline in total daily off-time (ie, return/worsening of motor/nonmotor symptoms often experienced between medication doses).

RESULTS After screening 824 adults with PD, 507 participants (mean [SD] age, 64.9 [8.5] years; 321 male [63%]; mean [SD] disease duration, 6.7 [4.5] years; mean [SD] baseline daily off-time, 5.5 [2.4] hours) were enrolled and received tavapadon (n = 252) or placebo (n = 255). Tavapadon significantly increased daily good-on-time by 1.10 hours compared with placebo (1.70 vs 0.60 hours; 95% CI, 0.60-1.70; P <.001). Daily off-time was significantly reduced from baseline with tavapadon vs placebo (−1.88 vs −0.93 hours; difference, −0.94 hours; 95% CI, −1.48 to −0.41]; P < .001). Tavapadon had a favorable safety profile; although more AEs occurred with tavapadon vs placebo (180 participants [71.7%] vs 140 participants [55.1%]), most were nonserious (93.2%) and mild to moderate in severity. Common AEs with tavapadon (≥5% of participants) were nausea (14.3%), dyskinesia (10.0%), and dizziness (7.6%).

CONCLUSIONS AND RELEVANCE Findings of this randomized clinical trial demonstrate the efficacy, tolerability, and safety profile of tavapadon adjunctive to levodopa for off fluctuations.

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Parkinson disease (PD) is a chronic, progressive, and disabling neurodegenerative disease impacting approximately 6.2 million people worldwide and resulting in 3.2 million disability-adjusted life-years.[1-3] Current treatments aim to ameliorate clinical symptoms by restoring dopamine activity in the brain.[1] Levodopa is standard of care for treatment of PD and provides rapid improvement of motor symptoms.[1,4] However, together with disease progression, long-term treatment with oral levodopa is complicated by motor and nonmotor fluctuations in up to 80% of patients.[1,4-8] These fluctuations are partially attributed to variability in drug absorption, progressive loss of striatal terminals that store, release, and reuptake dopamine, and the short plasma half-life of oral levodopa.[5,7,9] Motor complications can lead to substantial impairments, negative impacts on activities of daily living, and decreased quality of life,[6,10] all of which may be exacerbated by increasing doses of levodopa.[11] Survey data have shown that motor fluctuations are the most bothersome symptom of advanced PD.[12]

Other pharmacotherapies for PD (eg, dopamine agonists [DAs]) have been used as adjuncts to levodopa therapy to improve motor control by reducing off time and increasing on time without troublesome dyskinesia while avoiding complications associated with increasing doses of levodopa.[4,7,11,13] Currently available oral and transdermal DAs (eg, pramipexole, ropinirole, rotigotine) preferentially target D2/D3 receptors, and are associated with an increased risk of adverse events (AEs) compared with oral levodopa alone, which is hypothesized to be partially attributed to widespread distribution of D2/D3 receptors in the brain and peripheral nervous system.[4,14] These AEs include impulse control disorders (ICDs), hallucinations, somnolence, edema, and orthostatic hypotension, all of which may increase with higher doses and prolonged treatment.[4,14] Additionally, off-target activation of nondopaminergic receptors (eg, serotonin, norepinephrine, histamine) by currently available D2/D3 DAs has also been implicated in the development of motor and nonmotor AEs.[14]

Tavapadon is a novel, oral, once-daily, selective D1/D5 agonist.[15] It is hypothesized that tavapadon can improve motor control with once-daily dosing, and may reduce the risk of specific AEs associated with D2/D3 receptor activation observed with currently available D2/D3 DAs.[15] Phase 2 trials demonstrated improved motor function, with an acceptable tolerability and safety profile, in people with early PD and PD with motor fluctuations treated with flexible-dose tavapadon.[14,16,17] The Flexible-Dose, Adjunctive Therapy Trial in Adults With Parkinson Disease With Motor Fluctuations (TEMPO-3) was designed to evaluate the efficacy, safety, and tolerability of flexible-dose tavapadon (5-15 mg once daily) as an adjunctive therapy for PD in levodopa-treated adults experiencing off-time fluctuations.

Key Points

Question Can adjunctive tavapadon—an oral, once-daily, selective dopamine (D) D1/D5 agonist—improve motor control for people with Parkinson disease (PD) experiencing motor fluctuations while minimizing risk of adverse events?

Findings In this phase 3, double-blind, placebo-controlled, 27-week randomized clinical trial of 507 participants with PD, tavapadon significantly increased daily on-time without troublesome dyskinesia (good-on-time) vs placebo. Most adverse events were mild to moderate in severity with nausea, dyskinesia, and dizziness most common with tavapadon.

Meaning Results show that tavapadon adjunctive to levodopa provided clinically meaningful motor improvements and an acceptable safety profile in adults with PD experiencing motor fluctuations while receiving oral levodopa.

Methods

Study Design

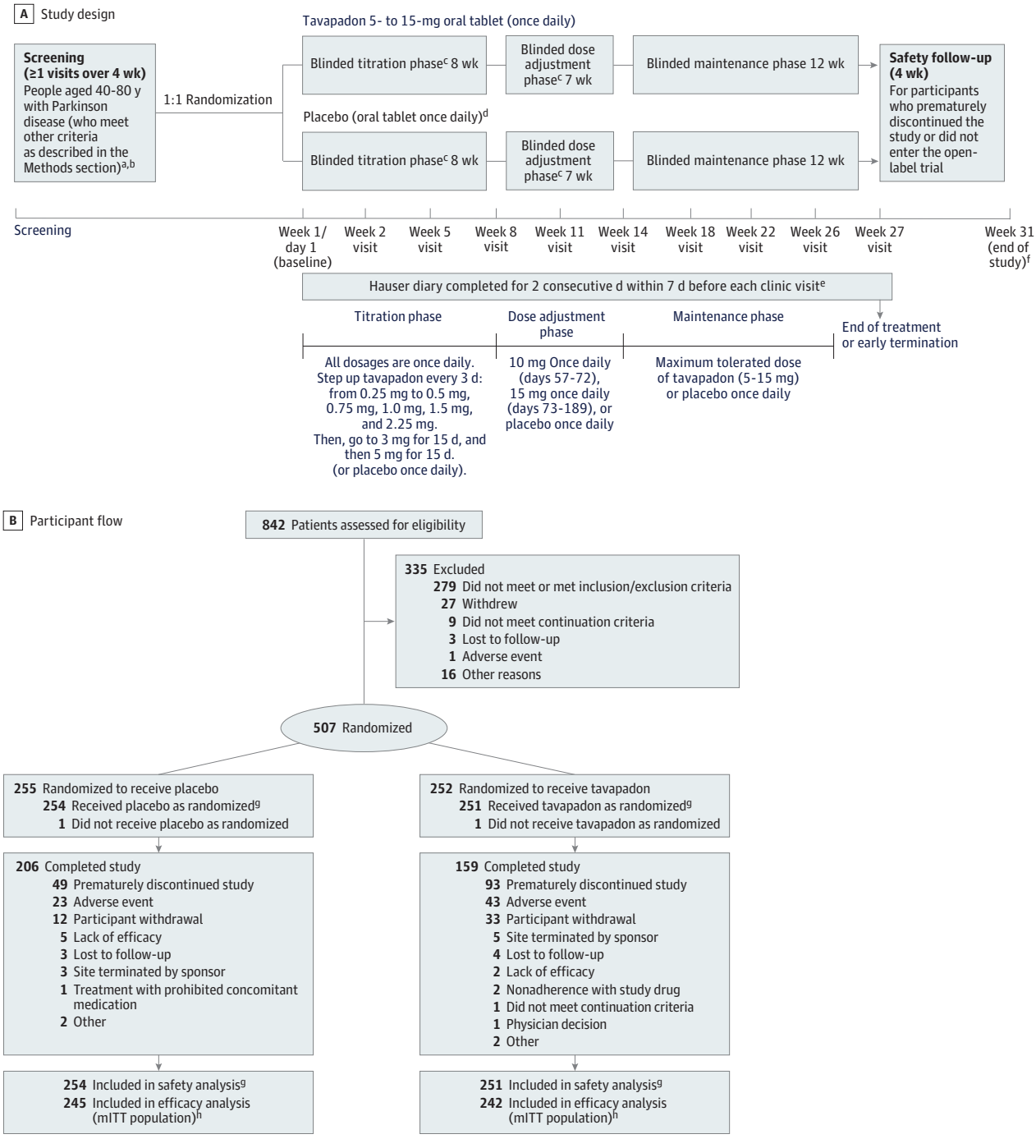
The TEMPO-3 trial was a prospective, phase 3, multicenter, double-blind, placebo-controlled, parallel-group, 27-week randomized clinical trial conducted at 148 clinical sites across 14 countries (the trial protocol and statistical analysis plan are available in Supplement 1 and Supplement 2, respectively). The trial was conducted from September 2020 to February 2024. Participation lasted approximately 35 weeks, including a 4-week maximum screening period, a 27-week treatment period, and a 4-week safety follow-up period (**Figure 1**A). The trial was conducted in accordance with the Declaration of Helsinki, International Council for Harmonisation Good Clinical Practice guidelines, and was approved by independent ethics committees before initiation; participants or their representatives provided written informed consent before screening and enrollment. This study followed the Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines.

Participants

Eligible participants included adults aged 40 to 80 years with a diagnosis of PD consistent with the UK Parkinson Disease Society Brain Bank criteria, a modified Hoehn & Yahr score of 2, 2.5, or 3 in the on-state (ie, good mobility, smoother movement, fewer motor/nonmotor symptoms), a minimum of 2.5 hours of daily off-time on 2 consecutive days during the screening period (ie, return/worsening of motor/nonmotor symptoms often experienced between medication doses), a minimum stable dose of 400 mg of levodopa per day divided at least 3 or 4 times daily, and a good response to levodopa per investigator judgment. Participants were required to demonstrate an acceptable ability to complete the Hauser diary[18] on concordance testing during the screening period, and to maintain the same levodopa dose and frequency for the trial duration. Exclusion criteria included atypical or secondary parkinsonian syndrome, previous or planned surgical intervention for PD, Montreal Cognitive Assessment score less than 26, history of or current diagnosis of a clinically significant impulse control disorder, history of psychosis or hallucinations for 12 months or less, or current use (≤14 days of baseline) of DAs. Previous and concurrent use of catechol-O-methyl transferase inhibitors, monoamine oxidase inhibitor B inhibitors, amantadine, istradefylline, or anticholinergic drugs was permitted if initiated more than 90 days before the baseline visit;

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Figure 1. Flowchart Depicting the TEMPO-3 Study Design and Participant Disposition



mITT indicates modified intention to treat; TEMPO-3, Flexible-Dose, Adjunctive Therapy Trial in Adults With Parkinson Disease With Motor Fluctuations.

[a]Diagnosis of Parkinson disease (PD) consistent with UK Parkinson Disease Brain Bank diagnostic criteria.

[b]Good response to levodopa and receiving a minimum stable dose of ≥400 mg per day (minimum 3-4 daily doses [≥3 doses of extended release levodopa/carbidopa or ≥4 doses of standard levodopa/carbidopa]).

[c]Tavapadon was titrated over 6 weeks to a 5-mg once-daily dose and blinded dose adjustment up to 15 mg once daily (maximum-tolerated dose) by week 14.

[d]Placebo treatment was on a background of levodopa.

[e]Primary end point was the change from baseline to end point in the total on time without troublesome dyskinesia (good-on-time) based on the 2-day average of the Hauser diary.

[f]Adverse events (AEs) and concomitant medications were recorded; nonserious AEs were followed up through week 31 and serious AEs were followed up to resolution.

[g]Safety analysis set.

[h]Efficacy analysis included the mITT population: all randomized participants who received at least 1 dose of study drug and had valid baseline and postbaseline Hauser diary assessments.

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dosage changes for concurrent PD medications were not permitted. Demographic information regarding race and ethnicity was included to accurately reflect the level of diversity of the study population and was self-reported by the participants. Available choices from which participants could select included American Indian or Alaska Native, Asian, Black, White, and other (not specified).

Randomization and Treatment

Participants were randomized 1:1 to receive flexible-dose tavapadon (5-15 mg once daily) or placebo as adjunct to levodopa, using a computer-generated randomization scheme. Participants were sequentially assigned to randomization numbers corresponding to study drug (tavapadon or placebo). All participants took 3 tablets of tavapadon, placebo, or a combination of tavapadon and placebo once daily during the treatment period; all tablets and packaging were identical in appearance for blinded dose adjustment. Tavapadon was gradually titrated to the maximum tolerated dose (up to 15 mg once daily) over the first 14 weeks of the trial during the dose titration and dose adjustment phases (eTable 1 in Supplement 3). Further adjustments during the maintenance phase were not permitted. For both treatment groups, the levodopa dose was fixed, without dose adjustments for the duration of the trial.

Outcomes

The primary end point was the change from baseline to week 26 in total daily on time without troublesome dyskinesia (referred to as *good-on-time)* based on the 2-day average of the Hauser diary.[19] The key secondary end point was change from baseline to week 26 in total daily off-time. Other secondary end points included change from baseline in good-on-time and off-time at all time points, change from baseline in Movement Disorder Society-Unified Parkinson Disorder Rating Scale (MDS-UPDRS) parts I, II, and III individual and combined scores (week 26 and all time points), Parkinson Disease Questionnaire (PDQ) 39 score,[20] and the EuroQol 5-Dimension 5-Level (EQ-5D-5L) index and visual analog scale (VAS) scores.[21] MDS-UDPRS assessments were conducted approximately 2 to 3 hours after the last dose of levodopa, and participant scores could have been collected during on or off status.

Change from baseline in MDS-UDPRS parts II+III scores was a predefined exploratory end point. Additionally, a post hoc analysis evaluated changes from baseline to week 26 (the end of treatment) in additional motor states (eg, on-time with non-troublesome dyskinesia, on-time without dyskinesia) across the normal waking day as recorded in the Hauser diary.

Safety end points included AEs, clinical laboratory values, vital signs (including orthostatic values), and electrocardiograms. Columbia-Suicide Severity Rating Scale (C-SSRS),[22] Epworth Sleepiness Scale (ESS),[23] and Questionnaire for Impulsive-Compulsive Disorders in Parkinson Disease-Rating Scale (QUIP-RS)[24] scores were also assessed at each visit.

Statistical Analysis

A sample size of 184 participants per group provided greater than or equal to 90% power to detect a 1-hour treatment difference in the primary outcome measure at a 2-sided .05 significance level, with an assumed SD of 2.5 hours. Because sample size assumptions were based on a preliminary trial with limited sample size and a different dosing schedule, an interim analysis to assess the adequacy of the sample size assumption was performed by an independent external review committee, and the target trial sample size subsequently increased from 368 to 500 participants to maintain statistical power at 90%. Participants, investigators, and all blinded sponsor personnel remained blinded until final database lock. Further statistical analysis details are included in the trial protocol (Supplement 1).

Efficacy analyses were conducted using a mixed model for repeated measures (MMRM) approach based on a modified intention-to-treat (mITT) population (ie, randomized participants who received ≥1 dose of the assigned treatment and had both baseline and ≥1 postbaseline Hauser diary entries). For the primary and key secondary analyses, MMRM included fixed effects of treatment, visits, and interaction between treatment and visit with baseline total daily on-time or off-time (for the primary and key secondary analysis, respectively) included as a covariate and participant included as a random effect. Statistical comparisons for other secondary end points were not corrected for multiplicity; therefore, *P* values should be interpreted as nominal. The safety analysis set included all participants who received 1 or more doses of tavapadon or placebo. Statistical analyses were performed with SAS, version 9.4 or higher (SAS Institute).

Results

Participants

Of 842 screened individuals, 507 (mean [SD] age, 64.9 [8.5] years; 186 female [37%]; 321 male [63%]) were randomized to receive tavapadon (n = 252) or placebo (n = 255) in addition to levodopa (randomization by site shown in eTable 2 in Supplement 3); 505 received at least 1 dose of study treatment and were included in the safety analysis, and 487 met criteria for the mITT population (Figure 1B). A total of 142 participants (28.0%) discontinued the trial, including 93 (36.9%) in the tavapadon group and 49 (19.2%) in the placebo group. Discontinuation reasons included both AEs and non-AE related reasons, such as withdrawal by participant and site terminations due to the war in Ukraine (Figure 1B). Demographics and baseline disease characteristics were similar between groups and reflective of a population with PD and motor fluctuations (**Table 1**); the mean (SD) duration since diagnosis was 6.7 (4.5) years, and participants experienced a mean (SD) of 5.52 (2.37) hours of daily off-time and 10.02 (2.60) hours of daily on-time without troublesome dyskinesia at baseline. Participants self-reported the following races and ethnicities: 2 American Indian or Alaska Native (0.4%), 4 Asian (0.8%), 6 Black (1.2%), 491 White (96.8%), and 4 other (0.8%).

Efficacy

The study met its primary end point, with the tavapadon group demonstrating a statistically significant increase of 1.1 hours in total daily good-on-time from baseline to week 26 compared

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Table 1. Baseline Demographics and Disease Characteristics

Characteristic	Placebo QD (n = 255)	Tavapadon 5-15 mg QD (n = 252)	Overall (N = 507)
Baseline demographics			
Age, mean (SD) [range], y	64.1 (8.5) [43-80]	65.6 (8.4) [43-80]	64.9 (8.5) [43-80]
Age group, No. (%), y			
<65	119 (46.7)	104 (41.3)	223 (44.0)
≥65	136 (53.3)	148 (58.7)	284 (56.0)
Sex, No. (%)			
Female	85 (33.3)	101 (40.1)	186 (36.7)
Male	170 (66.7)	151 (59.9)	321 (63.3)
Race, No. (%)			
American Indian or Alaska Native	2 (0.8)	0	2 (0.4)
Asian	2 (0.8)	2 (0.8)	4 (0.8)
Black	4 (1.6)	2 (0.8)	6 (1.2)
White	245 (96.1)	246 (97.6)	491 (96.8)
Other[a]	2 (0.8)	2 (0.8)	4 (0.8)
Weight, mean (SD), kg	79.9 (15.7)	79.5 (17.2)	79.7 (16.5)
BMI, mean (SD) [range][b]	27.0 (4.4) [16.6-40.2]	27.4 (5.1) [16.6-42.4]	27.2 (4.7) [16.2-46.4]
Baseline disease characteristics			
Time since initial diagnosis, mean (SD), y	6.4 (4.6)	6.9 (4.5)	6.7 (4.5)
Daily on-time without troublesome dyskinesia, mean (SD), h	10.1 (2.6)	9.9 (2.6)	10.0 (2.6)
Daily off-time, mean (SD), h	5.4 (2.5)	5.6 (2.2)	5.5 (2.4)
MoCA total score, mean (SD)	28.0 (1.3)	28.0 (1.3)	28.0 (1.3)
Modified H&Y stage, No. (%)[c]			
Stage 2	146 (57.3)	135 (53.6)	281 (55.4)
Stage 2.5	50 (19.6)	63 (25.0)	113 (22.3)
Stage 3	59 (23.1)	54 (21.4)	113 (22.3)
MDS-UPDRS part I score, mean (SD)[d]	7.5 (4.9)	8.0 (5.1)	7.7 (5.0)
MDS-UPDRS part II score, mean (SD)[d]	12.5 (7.1)	13.3 (6.5)	12.9 (6.8)
MDS-UPDRS part III score, mean (SD)[d]	32.4 (14.3)	32.6 (14.3)	32.5 (14.3)
MDS-UPDRS part II + III score, mean (SD)[d]	44.9 (18.9)	45.9 (18.2)	45.4 (18.5)
Use of adjunctive PD medication (added to levodopa regimen), No. (%)			
Yes	108 (42.4)	115 (45.6)	223 (44.0)
Amantadine	40 (15.7)	49 (19.4)	89 (17.6)
COMT inhibitors	19 (7.5)	17 (6.7)	36 (7.1)
Istradefylline	2 (0.8)	1 (0.4)	3 (0.6)
MAO-B inhibitors	77 (30.2)	76 (30.2)	153 (30.2)
No	147 (57.6)	137 (54.4)	284 (56.0)
Daily levodopa dose, mean (SD), mg	817.6 (644.0)	811.9 (451.4)	814.8 (556.1)

Abbreviations: BMI, body mass index; COMT, catechol-O-methyltransferase; H&Y, Hoehn & Yahr; MDS-UPDRS, Movement Disorder Society–Unified Parkinson Disease Rating Scale; MAO-B, monoamine oxidase B; MoCA, Montreal Cognitive Assessment; PD, Parkinson disease; QD, once daily.

[a] Race was self-reported by participants; these participants selected the category other to characterize their race.

[b] Calculated as weight in kilograms divided by height in meters squared.

[c] Evaluated in the on-state.

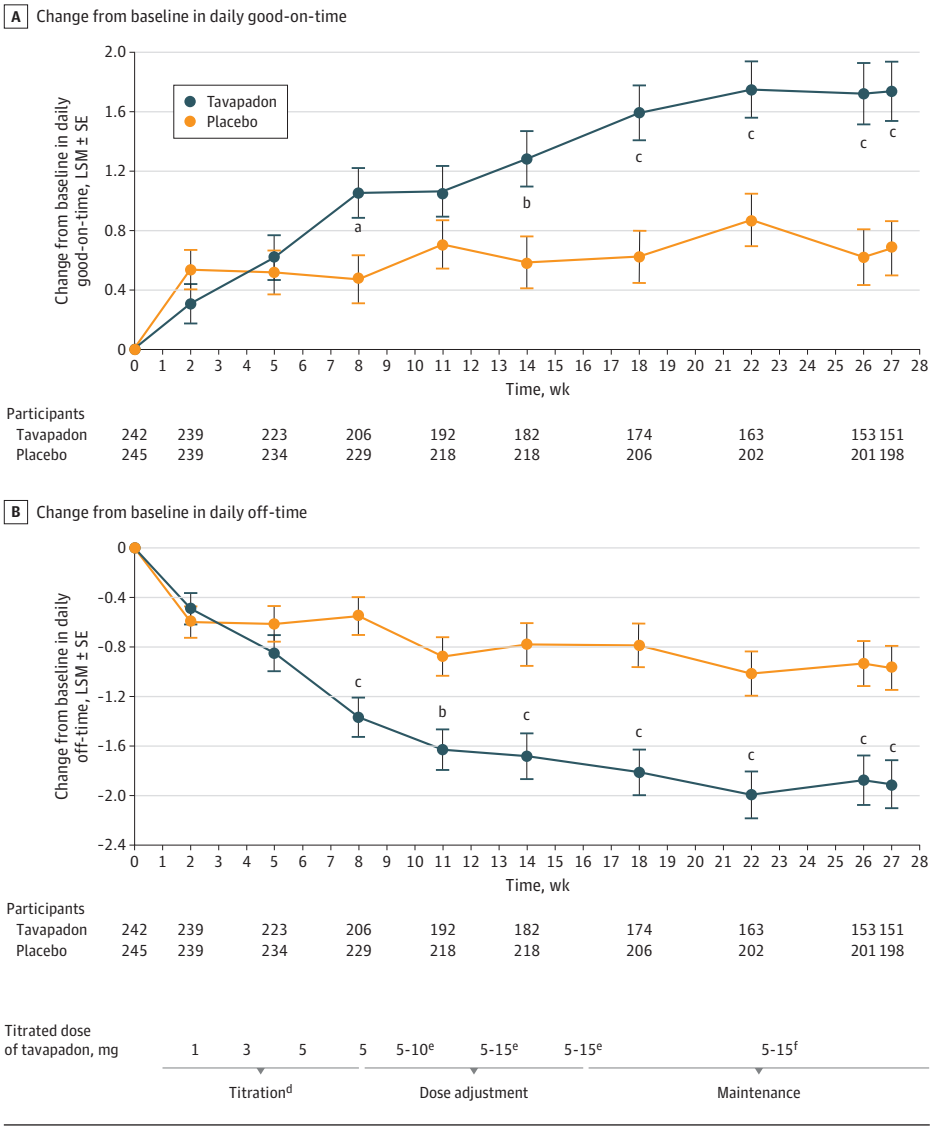
[d] Assessments conducted approximately 2 to 3 hours after last dose of levodopa and collected during on or off status.

with the placebo group (least-squares mean [LSM] increase of 1.70 hours with tavapadon vs 0.60 hours with placebo; LSM treatment difference, 1.1; 95% CI, 0.6-1.7; $P < .001$; Cohen $d = 0.40$). The increase in daily good-on-time from baseline was observed starting at week 2, and separation between tavapadon and placebo was observed as early as week 8 and sustained through week 26 and end of treatment (week 27) (**Figure 2**A). The tavapadon group experienced a statistically significant reduction from baseline to week 26 in the key secondary end point of off-time compared with the placebo group (LSM decrease of 1.88 vs 0.93 hours, respectively; LSM treatment difference, −0.94 hours; 95% CI, −1.48 to −0.41; $P < .001$; Cohen $d = 0.35$). Reductions in off-time with tavapadon relative to placebo were observed starting at week 8 and continued to end of treatment (Figure 2B). Predefined sensitivity analyses were consistent with the primary analysis for both primary and key secondary end points (data not shown).

Treatment with tavapadon also resulted in an improvement from baseline in MDS-UPDRS part II scores at week 26 compared with placebo (LSM decrease, 1.4 vs 0.1 points, respectively; LSM treatment difference, −1.2 points; 95% CI, −2.2 to −0.2; nominal $P = .02$), with significant reductions observed starting at week 11 and continuing to end of treatment (eFigure in Supplement 3). Reductions in MDS-UPDRS part III scores at

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Figure 2. Line Graphs Showing the Least-Squares Mean Change From Baseline in Daily On-Time (Hours) Without Troublesome Dyskinesia (Good-on-Time) and Off-Time in the mITT Population



Data based on the 2-day average of Hauser diary entries recorded at 30-minute intervals for 24 hours on 2 consecutive days. *Good-on-time* is defined as the sum of on time without dyskinesia and on-time with nontroublesome dyskinesia. Error bars are the SE. Nominal significance level of comparison with placebo without multiplicity adjustment except for primary time point. LSM indicates least-squares mean; mITT, modified intention to treat.

[a]$P < .05$.

[b]$P < .01$.

[c]$P < .001$.

[d]No deviations in the titration schedule up through 5 mg (week 8) were allowed.

[e]Dose adjustments were based on individual participant tolerability.

[f]Participants received the highest tolerated dose level that was achieved during the dose adjustment phase (5 mg, 22 participants [8.8%]; 10 mg, 32 participants [12.7%]; 15 mg, 169 participants [67.3%]). No adjustment in the tavapadon dose was allowed during the maintenance phase. Analysis was based on mixed models for repeated measures, with fixed effects for treatment group, visits, and treatment by visit interaction, and participant as a random effect.

week 26 were also significantly greater in participants treated with tavapadon vs placebo (LSM decrease, 7.0 vs 4.6 points, respectively; LSM treatment difference, −2.4 points; 95% CI, −4.3 to −0.5; nominal $P = .01$) (eFigure in Supplement 3). There were no notable differences in MDS-UPDRS part I, PDQ-39, or EQ-5D-5L Index and VAS scores (data not shown).

The exploratory analysis of change from baseline in MDS-UPDRS part II+III combined scores demonstrated a significant reduction in the tavapadon group vs the placebo group at week 26 (LSM decrease, 8.4 vs 4.7 points; LSM treatment difference, −3.7; 95% CI, −6.3 to −1.1; nominal $P = .005$) (eFigure in Supplement 3).

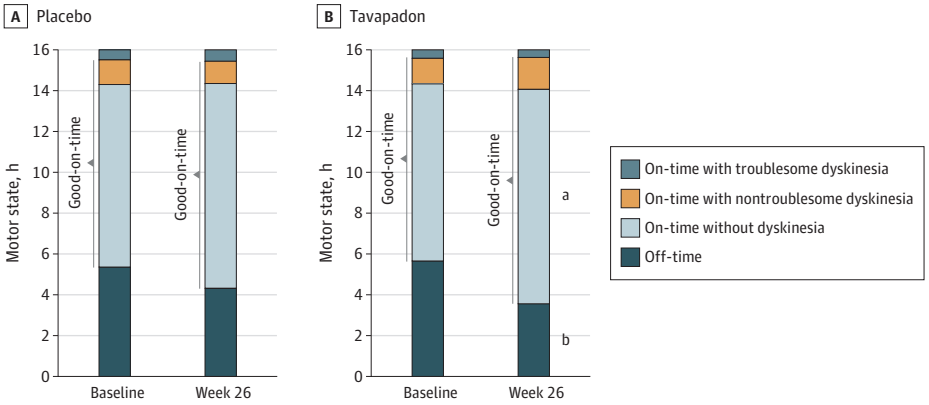
In a post hoc analysis, a comparison of changes from baseline in motor states across the waking day revealed the improvements in total daily good-on-time at week 26 were primarily driven by the increase in total daily on-time without dyskinesia (**Figure 3**).

Safety

In the safety population, 180 participants (71.1%) in the tavapadon group and 140 participants (55.1%) in the placebo group experienced at least 1 AE (**Table 2**). Most AEs were nonserious (93.2%) and mild to moderate in severity. Rates of serious AEs (Table 2) were similar between groups. Serious AEs reported in 2 or more participants in the tavapadon group were fall (n = 3) and abdominal pain (n = 2). Severe AEs observed in 2 or more participants in the tavapadon group were dyskinesia (3 [1.2%]), abdominal pain (3 [1.2%]), dizziness (2 [0.8%]), and nausea (2 [0.8%]). One AE of death due to metastatic pancreatic carcinoma reported in the tavapadon group was considered unrelated to study treatment. AEs led to treatment discontinuation in 17.1% of participants in the tavapadon group compared with 9.1% in placebo; discontinuations due to AEs occurred most commonly during the titration phase (placebo, 17 [7%]; tavapadon, 28 [11%]).

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Figure 3. Bar Graphs Depicting Hauser Diary Motor States Across the Normal Waking Day at Baseline and Week 26



Analysis of normal waking day was based on normalized waking hours of 16 hours per day for each individual patient. *Good-on-time* is defined as the sum of on time without dyskinesia and on-time with nontroublesome dyskinesia. *P* values indicate significant treatment difference in least-squares mean change from baseline to week 26.

[a]*P* < .05.

[b]*P* < .01.

Table 2. Summary of Adverse Events (Safety Population)

Safety outcomes	No. (%)	
	Placebo QD (n = 254)	Tavapadon 5-15 mg QD (n = 251)
AEs[a]	140 (55.1)	180 (71.7)
Mild	65 (25.6)	80 (31.9)
Moderate	64 (25.2)	82 (32.7)
Severe	11 (4.3)	18 (7.2)
AE related to study drug[b]	54 (21.3)	105 (41.8)
AEs by phase at onset, No./No. (%)		
Titration	83/254 (32.7)	100/251 (39.8)
Adjustment	54/237 (22.8)	100/214 (46.7)
Maintenance	65/220 (29.5)	71/188 (37.8)
Serious AEs	14 (5.5)	17 (6.8)
AE leading to discontinuation	23 (9.1)	43 (17.1)
Death	0	1 (0.4)[c]
Most common AEs (occurring with ≥5% incidence in tavapadon group), No. (%)		
Nausea	11 (4.3)	36 (14.3)
Dyskinesia	4 (1.6)	25 (10.0)
Dizziness	8 (3.1)	19 (7.6)
Headache	7 (2.8)	17 (6.8)
Fall	13 (5.1)	16 (6.4)
Orthostatic hypotension[d]	3 (1.2)	15 (6.0)
Hallucination, visual	3 (1.2)	14 (5.6)
COVID-19	9 (3.5)	13 (5.2)
Somnolence	11 (4.3)	13 (5.2)

Abbreviations: AE, adverse event; QD, once daily.

[a] Any event occurring after initiation of study drug.

[b] Assessed by investigator as being related to study drug.

[c] Participant withdrew from the study after receiving tavapadon for 35 days and died from metastatic pancreatic carcinoma on day 70 postrandomization. This death was unrelated to study drug.

[d] Defined as a decrease in systolic blood pressure greater than or equal to 20 mm Hg on standing compared with supine position or a decrease in diastolic blood pressure greater than or equal to 10 mm Hg on standing compared with supine position.

For the tavapadon group, the percentage of participants experiencing AEs was higher during the dose titration (39.8%) and adjustment (46.7%) phases compared with the maintenance (37.8%) phase. The most common AEs reported in the tavapadon group (≥5% incidence) were nausea (14.3%), dyskinesia (10.0%), dizziness (7.6%), headache (6.8%), falls (6.4%), and orthostatic hypotension (6.0%) (Table 2). Rates of nausea, dyskinesia, dizziness, and headache were higher during the titration phase and decreased with continued exposure. Rates of visual hallucinations and orthostatic hypotension were higher during the dose adjustment and maintenance phases compared with the titration phase.

Incidence rates of somnolence (tavapadon, 5.2%; placebo, 4.3%) and fatigue (tavapadon, 4.0%; placebo, 4.3%) were similar between treatment groups. Mean change from baseline to the last visit in ESS total score and subscores were also similar in the tavapadon and placebo groups, with a numerically lower percentage of participants reporting a score greater than or equal to 10 (sleepy) in the tavapadon groups vs the placebo group.

Of tavapadon-treated participants experiencing falls, only 1 was receiving treatment for orthostatic hypotension.

Overall, 9 participants (tavapadon, 6 [2.4%]; placebo, 3 [1.2%]) experienced AEs suggestive of ICDs. Of the 6 participants in the tavapadon group, all events were nonserious and mild to moderate in severity; 1 event led to tavapadon discontinuation. All events but 1 were resolved by the end of the trial. Mean changes from baseline to the last visit in QUIP-RS total scores were similar between the tavapadon and placebo groups (data not shown). There was no evidence of increased suicidality with tavapadon based on the C-SSRS (data not shown).

Blood pressure decreases in the tavapadon group were observed. At the end of the study (week 27), a mean (SD) decrease of 8.6 (16.3) mm Hg was observed in standing systolic blood pressure (vs 0.8 [15.2] mm Hg with placebo) and a mean (SD) decrease of 6.0 (10.8) mm Hg was observed in standing diastolic blood pressure (vs 1.8 [9.7] mm Hg with placebo), with similar changes observed for supine blood pressures. The frequency of measured orthostatic hypotension was comparable between treatment groups (eTable 3 in Supplement 3),

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but the incidence of orthostatic hypotension reported as an AE was higher in the tavapadon group (6.0%) compared with the placebo group (1.2%) (Table 2). There were no clinically relevant safety findings in clinical laboratory findings, eye examinations, or cardiovascular assessments.

Discussion

In this multinational, phase 3 randomized clinical trial, adjunctive tavapadon demonstrated clinically meaningful improvements in good-on-time in participants with PD experiencing off fluctuations despite optimized levodopa. Once-daily, flexible-dose (5-15 mg), oral tavapadon adjunctive to levodopa increased daily good on-time while significantly reducing daily off-time. Improvements in good-on-time are associated with better clinical outcomes, improved quality of life, and reduced economic burden.[25] The LSM decrease in off-time from baseline to week 26 observed in the tavapadon group (−1.88 hours) is comparable with improvements in off-time seen with other adjunctive PD therapies, including D2/D3 agonists, and surpasses the previously reported minimum clinically important difference for a reduction in off-time of approximately 1.0 hour.[26-28] The increase in daily good-on-time was specifically attributable to an increase in daily on-time without dyskinesia. Finally, participants treated with tavapadon showed nominally significantly greater reductions in MDS-UPDRS parts II and III scores, corresponding to improvements in activities of daily living and investigators' evaluations of motor function, respectively. No significant differences between groups were observed in the PDQ-39, EQ-5D-5L index, and VAS scores.

The safety profile of tavapadon was favorable and consistent with earlier-phase tavapadon trials[16,29]; most AEs were nonserious, mild or moderate in severity, and allowed participants to remain on treatment. Rates of discontinuation due to AEs were higher for tavapadon vs placebo. Common dopaminergic AEs (eg, nausea, dyskinesia, dizziness) reported in participants treated with tavapadon were expected.[4,7] Although decreases in blood pressure were observed in tavapadon-treated participants, cardiovascular parameters, including measured orthostatic hypotension and electrocardiogram abnormalities, were notably comparable between treatment groups, without clinically meaningful differences. In contrast, cardiovascular hypotension hindered the clinical development of full D1/D5 DAs (eg, ABT-431 and dihydrexidine). Falls incidence was comparable between tavapadon- and placebo-treated participants.

Currently available DAs preferentially target D2/D3 receptors and are associated with increased risk of nonmotor AEs, including ICDs and somnolence.[14,15] In this 27-week trial, there were no notable differences between tavapadon and placebo in the incidence of ICDs or the QUIP-RS total score. Treatment with tavapadon also did not increase the risk of somnolence based on both incidence of AEs of somnolence or fatigue and mean changes from baseline in ESS scores. As some AEs associated with dopaminergic treatment, such as ICDs, may emerge with prolonged treatment, additional data from the long-term extension trial[30] of tavapadon may further the understanding about the safety of tavapadon adjunctive to levodopa in people with PD and motor fluctuations.[14,15]

Strengths and Limitations

The main strength of this study was its rigorous multicenter, double-blind, randomized design. The 27-week treatment duration was comparable with other phase 3 trials investigating therapies for PD with motor fluctuations[31,32]; however, as people with PD in real-world settings may receive treatment for years,[33] research is needed to support the long-term safety and efficacy of tavapadon. This study was not designed to compare tavapadon with D2/D3 DAs or other adjunctive therapies for PD. The study population was predominantly White (96.8%), and further study of tavapadon in underrepresented populations may be needed. This trial was held during the height of the COVID-19 pandemic, which may have contributed to the longer-than-usual times to complete enrollment, and the discontinuation rate.

Conclusions

Results of this randomized clinical trial show that tavapadon, a novel, selective, D1/D5 agonist, provided clinically meaningful and sustained benefit with once-daily dosing adjunctive to levodopa in people with PD experiencing off fluctuations. Tavapadon has a favorable safety profile, which is consistent with findings from previous trials of tavapadon.[16,29] These findings highlight the clinical utility of adjunctive tavapadon for the treatment of PD with off fluctuations.

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